

Remark 9.2. We point out that one can also define analogues of these operators in the quasi-parabolic descendent algebras $\mathbb{D}_{\alpha_\bullet}^{\text{qpar}}$; recall that this is a supercommutative algebras generated by symbols $\text{ch}_k(\gamma)$ for $\gamma \in H^*(C)$ and $\text{ch}_k(e_j)$ for $0 \leq j \leq l+1$, modulo certain simple relations; alternatively, it is generated by $\text{ch}_k^H(\gamma), \text{ch}_k^H(e_j)$. One defines $\mathbb{L}_n^{\text{qpar}}: \mathbb{D}_{\alpha_\bullet}^{\text{qpar}} \rightarrow \mathbb{D}_{\alpha_\bullet}^{\text{qpar}}$ by $\mathbb{L}_n^{\text{qpar}} = \mathbb{R}_n^{\text{qpar}} + \mathbb{T}_n^{\text{qpar}}$ where $\mathbb{R}_n^{\text{qpar}}$ is defined exactly in the same way as in the parabolic case, and $\mathbb{T}_n$ is

$$\mathbb{T}_n^{\text{qpar}} = \sum_{a+b=n} a!b! \left((1-g)\text{ch}_a(\text{pt})\text{ch}_b(\text{pt}) + \sum_{j=0}^l \text{ch}_a(e_j)\text{ch}_b(e_{j+1}-e_j) \right).$$

In particular, the Virasoro constraints do not depend on a choice of weights within a fixed chamber.

The main result of this section are the Virasoro constraints for moduli spaces of parabolic bundles:

Theorem 9.3. Let $\alpha = (r, d, f_\bullet, c) \in C(I)$. For every $D \in \mathbb{D}^{\text{par}}$ we have

$$\int_{[M_\alpha]} \mathbb{L}_{\text{wt}_0}(D) = 0.$$

These generalize the Virasoro constraints for moduli spaces of stable bundles shown in [BLM]. For the proof, we will need compatibility statements between the Virasoro constraints and the structures that we have discussed earlier in the paper: wall-crossing, flag bundles, and Hecke operators. We develop these compatibilities first, and then give the proof in Section 9.5.

9.1. Primary states and wall-crossing compatibility. Compatibility with wall-crossing is already the key ingredient in [BLM]. To state it, let us introduce the Lie subalgebra of primary states.

Definition 9.4. The space of primary states of weight i on $\mathbf{V}_{\text{tr}}^{\text{par}}$ is

$$P_i = \{v \in \mathbf{V}_{\text{tr}}^{\text{par}} : L_0(v) = iv \text{ and } L_n(v) = 0 \text{ for } n > 0\}.$$

The space of primary states on the Lie algebra $\check{\mathbf{V}}_{\text{tr}}^{\text{par}}$ is

$$\check{P}_0 = P_1/T(P_0) \subseteq \check{\mathbf{V}}_{\text{tr}}^{\text{par}}.$$

By [LM, Corollary 5.7], a class $u \in \check{\mathbf{V}}_{\text{tr}}^{\text{par}}$ (of non-trivial topological type) is a primary state if and only if $L_{\text{wt}_0}(u) = 0$, where

$$L_{\text{wt}_0} = \sum_{n \geq -1} \frac{(-1)^n}{(n+1)!} L_{-1}^{n+1} \circ L_n : \check{\mathbf{V}}_{\text{tr}}^{\text{par}} \rightarrow \mathbf{V}_{\text{tr}}^{\text{par}}$$

is the dual of $\mathbb{L}_{\text{wt}_0}$.¹⁷ In particular, the moduli space M_α satisfies the Virasoro constraints (i.e. Theorem 9.3 holds) if and only if $[M_\alpha] \in \check{P}_0$ is a primary state.

We recall the reader that $\mathbf{V}_{\text{tr}}^{\text{par}}$ is a lattice vertex algebra (cf. Proposition 3.1). When the symmetrized Euler pairing $\chi_{\mathcal{P}}^{\text{sym}}$ is non-degenerate, this vertex algebra

¹⁷When $\mathbf{V}_{\text{tr}}^{\text{par}}$ admits a conformal element ω , L_{wt_0} is a canonical lift of the operator $[-, \omega]$, see [BLM, Proposition 3.13, Lemma 3.15].